Supplemental Material for
"Paternal age in rhesus macaques is positively associated with germline mutation accumulation but not with measures of offspring sociability"

Short title: Mutation rate in rhesus macaques
Keywords: mutation rate; paternal age effect; neurodevelopmental disorder

Supplemental Results

Elevated transition rates in rhesus macaque

C>T transitions accounted for 54% of all observed *de novo* mutations, significantly higher than the 42% found in human (χ^2 test, $p = 1.5 \times 10^{-4}$; Jónsson et al. 2017). This excess in C>T transitions results in a higher transition-to-transversion ratio in macaque (2.84) than observed in humans (2.22). This excess also contributes to a higher mutation rate toward weak base pairs (A:T) versus strong base pairs (G:C) when compared to humans (macaque: 3.51, human: 2.11). A possible explanation for the higher transition-to-transversion ratio than expected from substitution rates measured between species (Moorjani et al. 2016) may be a lower parental age among macaques in our sample (7.5 years), relative to the estimated median in nature (11 years; Xue et al. 2016). Parental age in humans, particularly increasing maternal age, increases the proportion of C>G transversions while decreasing the proportion of CpG>TpG transitions (Jónsson et al. 2017; Gao et al. 2019).

Supplemental Methods

Sequencing library preparation

Standard PCR-free Illumina paired-end sequencing libraries were generated using KAPA Hyper PCR-free library reagents (KK8505, KAPA Biosystems) in Beckman robotic workstations (Biomek FX and FXp models). We sheared total genomic DNA (500 ng) into fragments of approximately 200-600 bp using the Covaris E220 system (96-well format) followed by purification of the fragmented DNA using AMPure XP beads. We next employed a double size selection step with different ratios of AMPure XP beads to select a narrow size band of sheared DNA molecules for library preparation. DNA end-repair and 3′-adenylation were performed in the same reaction followed by ligation of the barcoded adaptors to create PCR-Free libraries. We ran each library on the Fragment Analyzer (Advanced Analytical Technologies, Ames, Iowa) to assess library size and presence of remaining adaptor dimers. This was followed by qPCR assay using KAPA Library Quantification Kit and their SYBR FAST qPCR Master Mix to estimate the size and quantification. These WGS libraries were sequenced on an Illumina HiSeq X instrument to generate 150 bp paired-end reads. All flow cell data (BCL files) were converted to barcoded FASTQ files.

Variant calling

Single nucleotide variants and small indels (up to 60 bp) were called using GATK version 3.6 following best practices. We used HaplotypeCaller to generate gVCFs for each sample, and performed joint genotype calling on all samples using GenotypeGVCFs, generating a VCF file. GATK hard filters (SNPs: “QD < 2.0 || FS > 60.0 || MQ < 40.0 ||

MQRankSum < -12.5 || ReadPosRankSum < -8.0"; Indels: "QD < 2.0 || FS > 200.0 || ReadPosRankSum < -20.0") were applied and we removed calls that failed.

We also ran FreeBayes on the same BAM files as an alternative pipeline for variant calling. Following the recommended procedure, we applied the following filters on the resulting calls: ("QUAL > 1 & QUAL / AO > 10 & SAF > 0 & SAR > 0 & RPR > 1 & RPL > 1").

Estimating the fraction of callable sites

The callability we calculate for each trio is used as an estimate of, and correction for, false negatives. Our calculation differs slightly from Besenbacher et al. (2019) in that it combines α_{site}, the estimate of false positives, and $\Sigma C_f^{de\ novo}$ with a sampling approach. We altered the approach out of concern that the calculation of α_{site} relies on the assumption of independence between multiple filtering tests. To deal with this, we applied a sampling approach to estimate these values simultaneously from the set of sites that pass our filters in a single measure of callability, ΣC_i. Our correction is estimated for each individual, C_c, C_p, and C_m, from a random sample of 250,000 sites across the genome that match the filtering criteria. Though this continues to rely on the assumption of independence between the callability of each individual in the trio, we believe our approach makes fewer assumptions about the joint probabilities of site filtering.

Poisson regression

We estimated the effect of parental age on the mutation rate with a Poisson regression, modeling the number of mutations for trio *i* as:

$$N_{\text{mut},i} \sim \text{Pois}(\mu_{g,i} N_{\text{sites},i})$$

where $\mu_{g,i}$ is the per-generation mutation rate in trio i and $N_{\text{sites},i} = 2 \cdot \sum_x C_i(x)$, the diploid callable genome size for trio i. We used an identity link in the Poisson regression, which fit better than the canonical link (lower AIC value), with:

$$\mu_{g,i} = \beta_0 + \beta_1 X_{p,i} + \beta_2 X_{p,i}$$

where X_p and X_m are the paternal and maternal ages for trio i, and respective regression coefficients β. To adjust for differences in observable genome size, we generated two new variables:

$$Z_{p,i} = N_{\text{sites},i} X_{p,i} \qquad \text{and} \qquad Z_{m,i} = N_{\text{sites},i} X_{m,i}$$

and fit the Poisson regression model on $N_{\text{sites},i}$, $Z_{p,i}$, and $Z_{m,i}$ with no intercept. That is:

$$\mu_{s,i} = \beta_N N_{\text{sites},i} + \beta_p Z_{p,i} + \beta_m Z_{m,i}$$

where $\mu_{s,i}$ is the per-site per-generation mutation rate in trio i.

In this regression model, the coefficient for $N_{\text{sites},i}$ represents the per-site version of the original intercept β_0, and the coefficients for $Z_{p,i}$ and $Z_{m,i}$ are the effects of the respective parental age on the per-site per-generation mutation rate.

We used an unequal variances t-test to detect differences in the age effect on mutation rates between species. The test statistic was equal to the absolute difference between coefficients divided by the pooled standard error. We calculated the power of this test to detect differences in the age effect between species by simulating macaque mutations in each trio under the Poisson model with different β_P coefficients. In 10,000 simulations with $\beta_P = 4.27 \times 10^{-10}$, we could only detect a difference from the human coefficient 27% of the time ($p < 0.05$ level). At the 95% CI upper bound for β_P in the macaque, we were able to detect a difference from the human coefficient 87% of the time.

Personality characteristic scale

Observer rated each animal using a 7-point Likert-type scale on three trait adjectives. Previous work (Capitanio and Widaman 2005) had demonstrated these ratings form a scale that reflected the personality characteristics Sociability. The component adjectives were: Sociability—affiliative, agreeable, sociable, appears to like the company of others and seeks out social contact with other animals; Warm—seeks or elicits bodily closeness, touching, grooming; and Solitary—subject prefers to spend considerable time alone, avoids or does not often seek contact with other animals. Observers were trained to show greater than 85% agreement on coding behavior, and Cronbach's alpha (a measure of scale reliability) for the Sociability scale was 0.92.

Supplemental Figures

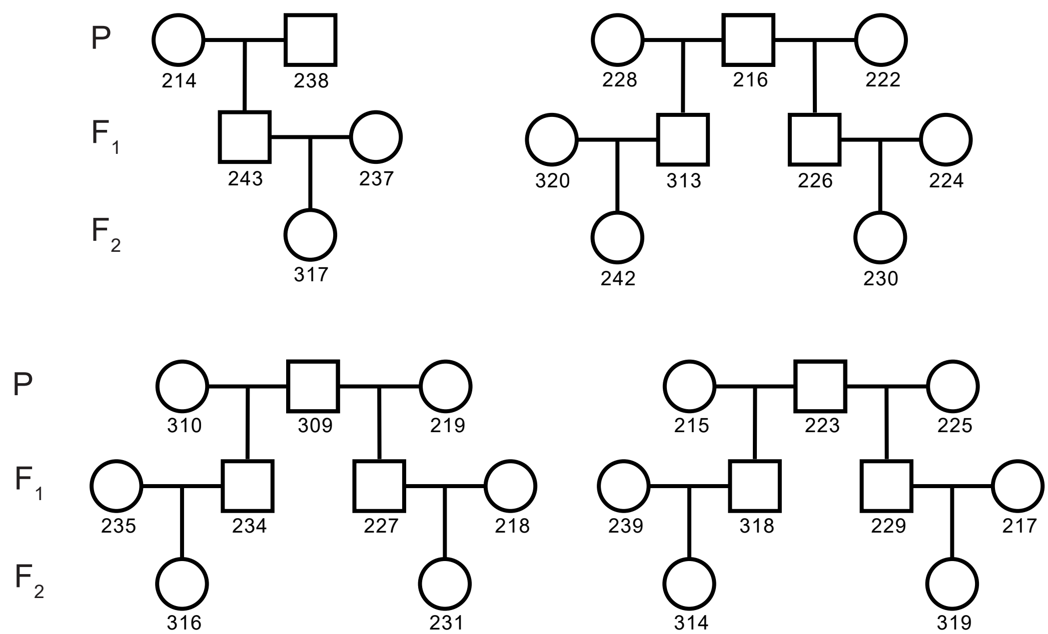

Figure S1. Pedigree structures of sequenced macaques

The 32 individuals sampled in this study were each part of the three-generation families depicted above.

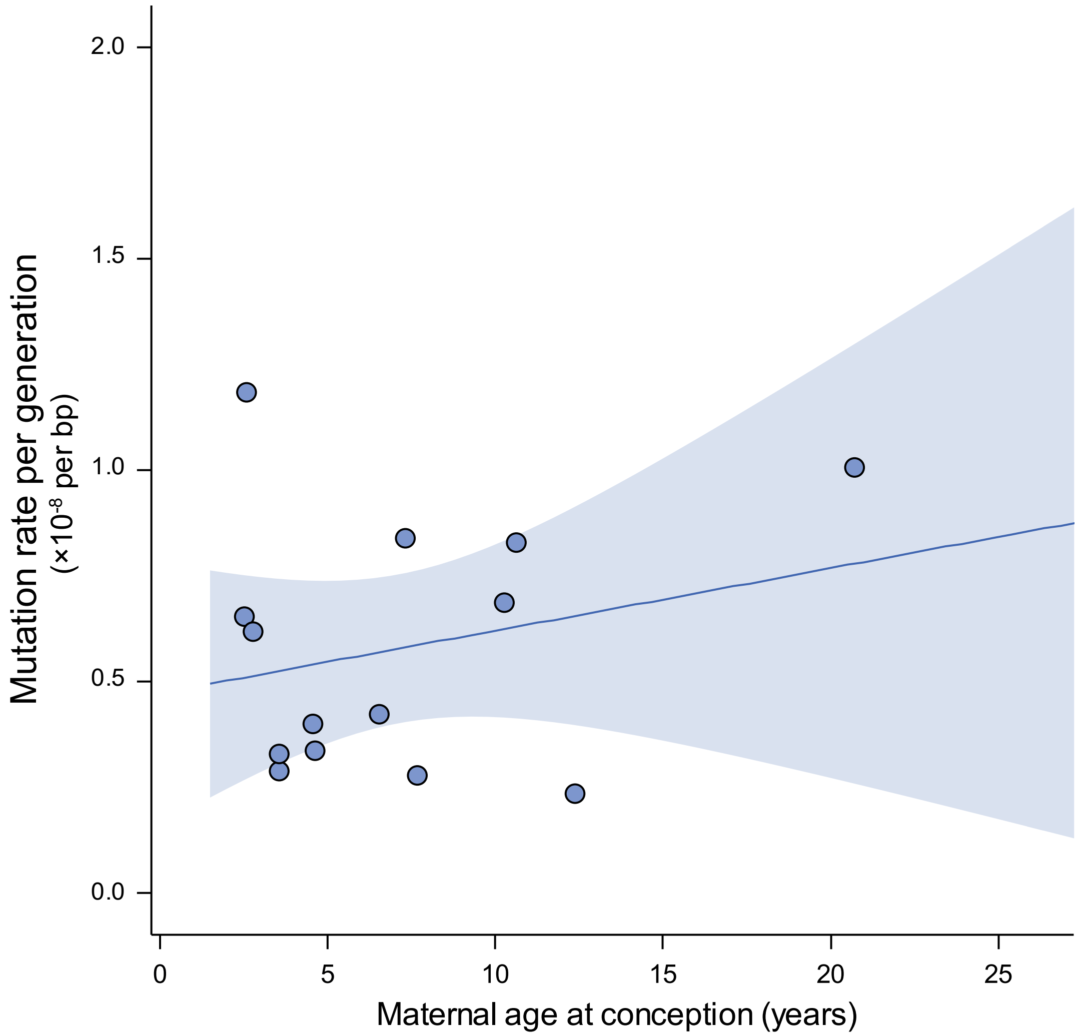

Figure S2. Effect of maternal age on mutation rate

Estimates of the overall mutation rate from 14 independent rhesus macaque trios with respect to the maternal age at conception. No significant relationship exists between overall mutation rate and maternal age at conception ($R^2 = 0.02$, $p = 0.39$; shaded area shows 95% CI).

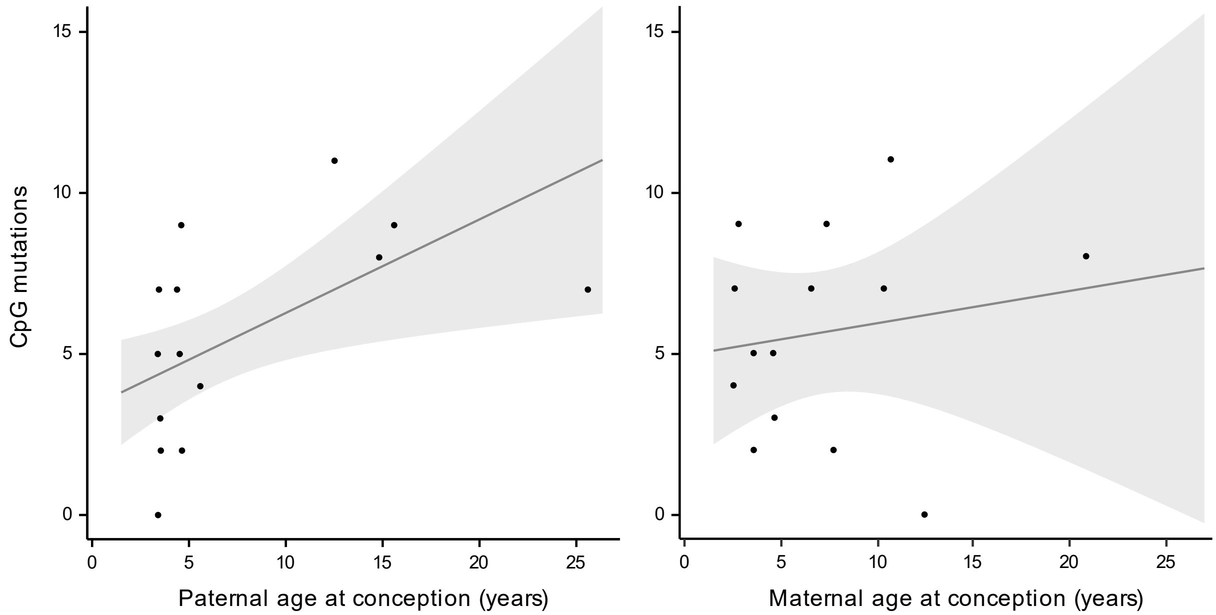

Figure S3. Mutations at CpG sites and parental age

Mutations at CpG sites with respect to the paternal (left) and maternal (right) age in each of the 14 trios. Shaded area shows regression 95% CI. As with the overall mutation rate, there is a significant relationship between the CpG mutation rate and paternal age ($R^2 = 0.27$; Poisson regression, $p = 0.015$) while we find no evidence for a relationship with maternal age ($R^2 = 0.03$; Poisson regression, $p = 0.94$).

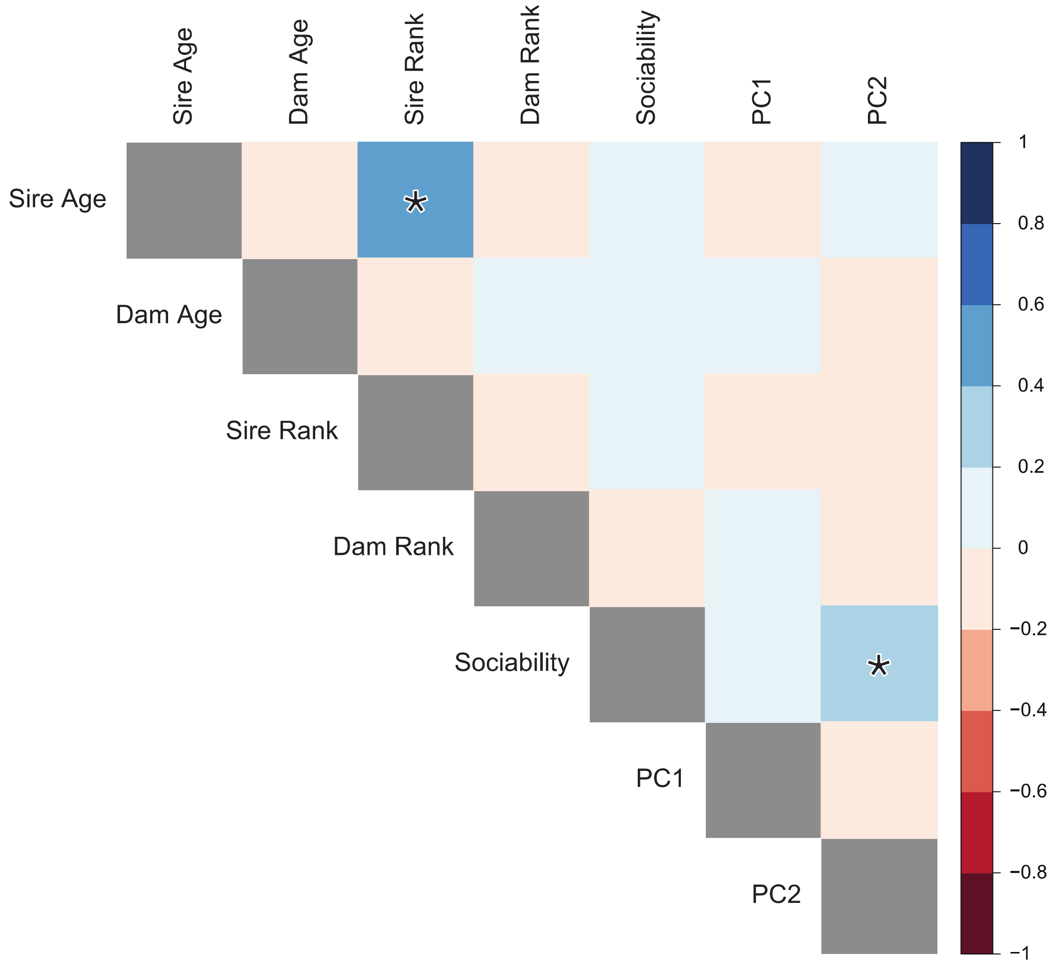

Figure S4. Correlations between parental age and behavioral traits in male rhesus monkeys. Boxes are shaded by the intensity of correlation in pairwise comparisons between each row and column. Diagonal values have been omitted. Legend on the right shows range of Pearson's correlation coefficient for each color. Significant correlations ($p < 0.05$) are highlighted with an asterisk.

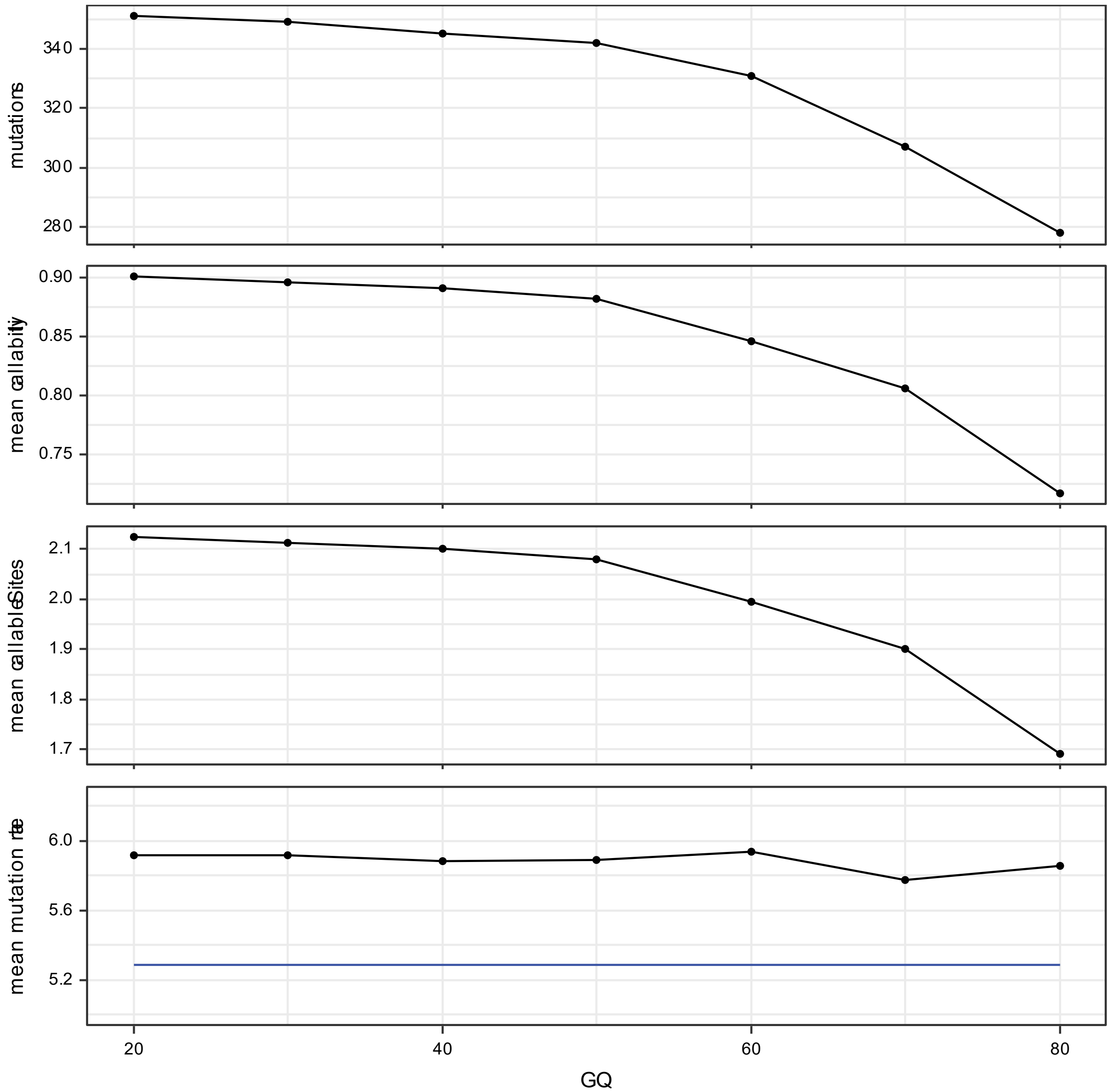

Figure S5. Filtering GATK candidate mutations by genotype quality

Plots show the effect of increasingly stringent GQ filters on the number of candidate mutations, mean callability, mean number of callable sites ($\times 10^9$ bp), and the mean mutation rate ($\times 10^{-8}$ per bp, per generation) across all trios. The mean mutation rate estimated from the *freebayes* calls is shown in blue for comparison.

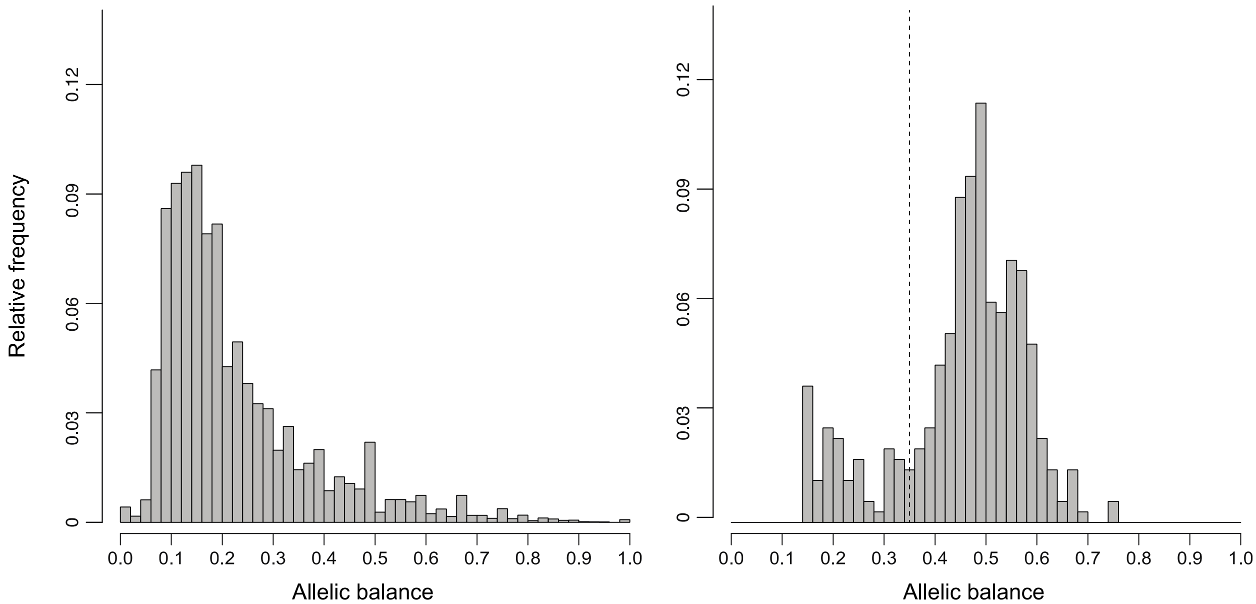

Figure S6. Allelic balance among candidate mutations

Distribution of allelic balance in heterozygote offspring that are Mendelian violations (left) and distribution at candidate *de novo* mutations after applying filters (right). Allelic balance at sites calculated as depth of alternate allele divided by total depth; dashed line indicates cutoff of 35% alternate depth.

Table S1. Loadings for first four principal components of behavioral observations

Parameter[a]	PC1	PC2	PC3	PC4
API(M)	-0.02	0.02	-0.01	-0.02
API(F)	0.02	0.01	0.03	-0.02
CON(M)	-0.30	0.49	0.01	0.65
CON(F)	0.44	0.31	-0.81	0.10
PRX(M)	-0.46	0.66	0.01	-0.51
PRX(F)	0.70	0.46	0.54	-0.04
GRI(M)	-0.08	0.12	-0.02	0.22
GRI(F)	0.07	0.04	-0.20	-0.51

[a] Abbreviations,
API approaches initiated to males(M) and females(F)
CON contact with males and females
PRX proximity to males and females
GRI grooming initiated to males and females

Table S2. Partial correlation of offspring sociability and sire age while controlling for sire rank

Measure of social function	Pearson's *r*	*p*-value
Sociability	0.06	0.49
Behavioral PC1	-0.08	0.31
Behavioral PC2	0.04	0.64
Behavioral PC3	-0.07	0.37
Behavioral PC4	0.09	0.27

References

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